

ESP32-S3-Based TinyML Framework with FMCW Radar and Event-Driven Vision for Low-Power Teaching Activity Recognition

Abstract

Teacher absenteeism and weak administrative oversight remain major obstacles for rural education systems, yet continuous video surveillance is rarely practical due to power, bandwidth and privacy constraints. This work presents an edge AI-based teaching activity monitoring framework built entirely on an ESP32-S3 microcontroller, designed to operate within the limited electrical infrastructure of rural classrooms while preserving visual privacy. The architecture uses a hierarchical, event-driven sensor fusion strategy: a low-power frequency-modulated continuous-wave (FMCW) radar continuously tracks micro-motion to verify presence, while a higher-power CMOS camera is activated only when spatial and acoustic thresholds are satisfied. This “check-and-sleep” logic minimizes camera duty cycle and shifts the baseline sensing burden onto the radar layer. A proprietary Teaching Activity Recognition (TAR) dataset was collected from non-uniform rural classrooms, covering four activity classes (standing, writing, sitting and vacant) under varied lighting and exposure conditions. The visual inference pipeline deploys a fully INT8-quantized MobileNet model via TensorFlow Lite Micro on the ESP32-S3, with JPEG payloads streamed into PSRAM to avoid CPU stalls during inference. Empirical evaluation shows that the edge node achieves 89.0% classification accuracy across the four activity classes, while analytical modelling and field measurements indicate an annualized energy consumption below 1.45 kWh per classroom under realistic duty cycles. By processing raw optical data locally and transmitting only compact activity labels to the backend, the proposed node offers a practical, privacy-conscious and energy-efficient engineering solution for scalable rural teaching activity monitoring and governance.

Keywords: Edge Computing, TinyML, ESP32-S3, Teaching Activity Recognition (TAR), Sensor Fusion, Green IoT

1. Introduction

Education serves as the primary catalyst for economic mobility; however, systemic inefficiencies persistently plague the educational sectors of developing nations [1]. A critical, localized challenge is unmonitored teacher absenteeism, which fundamentally degrades student literacy and retention metrics. In countries such as Bangladesh, logistical and financial constraints prevent consistent administrative supervision in rural schools. This creates a severe “governance gap,” leaving classrooms completely unsupervised for extended durations [2].

While technological interventions offer a theoretical solution, current Human Activity Recognition (HAR) frameworks typically rely on continuous, cloud-dependent video surveillance [3, 4]. Such centralized approaches, however, present three significant barriers for deployment in the Global South. First, the prohibitive resource demands of continuous video streaming—necessitating high-bandwidth connectivity and constant power—are often unattainable in rural infrastructure. Second, the inherent privacy intrusion of “always-on” optical surveillance fosters significant institutional resistance by compromising the privacy of educators and students alike. Third, the computational intensity of standard Convolutional Neural Networks (CNNs) far exceeds the SRAM and processing thresholds of affordable,

low-tier microcontrollers.

This framework is fundamentally motivated by a critical operational asymmetry: while classrooms remain occupied for extended durations, the behavioral shifts that signify instructional activity occur only over sparse, localized intervals. Consequently, continuous passive monitoring via power-intensive vision networks introduces significant mathematical and energetic redundancies. Recognizing that macroscopic presence verification can be decoupled from granular visual classification, the proposed architecture shifts the primary baseline sensory burden away from the vision stack onto an ultra-low-power, non-invasive medium.

To bridge this gap, we present a sustainable Edge AI monitoring framework deployed on the ultra-low-cost ESP32-S3 System-on-Chip (SoC) [5, 6]. Moving away from passive duty-cycling, our design introduces an event-driven “Check-and-Sleep” topology. By establishing a tiered-sensing logic, the system utilizes a millimeter-wave (mmWave) FMCW radar to continuously scan for micro-Doppler signatures while keeping the high-power vision pipeline in a native deep-sleep state [7, 8].

The core algorithmic contribution of this paper is the successful deployment of a Full INT8-quantized MobileNet classification pipeline directly onto the MCU’s Pseudo-Static RAM (PSRAM). By fusing this highly compressed neural network with our autonomous “SD-Watchdog” net-

work failover mechanism, the proposed node achieves an exceptional 89.0% detection accuracy while mathematically capping its annualized energy draw at under 1.5 kWh [9, 10].

In summary, this work makes the following engineering contributions:

- We design an ESP32-S3-based edge node that fuses FMCW radar, CMOS imaging and acoustic thresholds in an event-driven pipeline, enabling real-time teaching-activity recognition under strict energy and hardware constraints.
- We deploy a fully INT8-quantized MobileNet on the microcontroller using PSRAM-backed buffering, and characterize its memory footprint and inference latency to demonstrate feasibility of vision-based TinyML on low-cost rural infrastructure.
- We model and empirically measure the annualized energy consumption of the complete node under realistic classroom duty cycles, showing that continuous monitoring can be sustained at under 1.5 kWh per year and is therefore practical for weak grids.
- We implement a privacy-preserving architecture in which raw optical data is processed locally and discarded immediately, transmitting only compact activity classification outputs to the governance backend, thereby reducing both bandwidth demand and exposure of sensitive classroom imagery.

2. Related Work

The paradigm of Machine Learning within the Internet of Things (IoT) is actively shifting from centralized cloud execution toward decentralized, energy-efficient edge intelligence.

Traditional monitoring frameworks [11, 12, 13] depend on cloud-centric topologies where raw sensor telemetry is streamed to remote servers. While this enables the utilization of expansive architectures such as Faster R-CNN [14], it introduces fatal latency issues and massive bandwidth dependencies. Furthermore, continuously streaming classroom environments over insecure networks presents profound privacy vulnerabilities [15, 16]. Recent literature strongly advocates for Edge AI [17, 18], ensuring that raw, sensitive optical data is processed locally and immediately discarded, transmitting only the finalized classification vector.

Executing vision-based recognition on silicon with sub-megabyte SRAM is the fundamental challenge of TinyML [19]. Frameworks like TensorFlow Lite Micro (TFLM) facilitate the deployment of optimized models via aggressive pruning and quantization [20, 21]. While YOLO variants (v4, v8) dominate high-accuracy behavior detection on edge GPUs [19], they remain computationally unviable for IoT microprocessors. Consequently, research is pivoting toward depthwise separable convolutions (e.g., MobileNet,

ShuffleNet) that offer high spatial efficiency [22]. A comprehensive summary of contemporary literature, detailing target domains, hardware deployments, and underlying model architectures, is provided in Table 1

Continuous duty-cycling invariably wastes power during empty classroom periods [3]. Contemporary methodologies favor event-driven power gating [14, 23]. By leveraging ultra-low-power sensing modalities like FMCW Radar to continuously monitor spatial displacement, energy-intensive image processing is relegated to a "last resort" operation, dramatically extending hardware longevity [24, 25].

3. Data Collection and Experimental Environment

To robustly train and evaluate the proposed architecture, a proprietary Teaching Activity Recognition (TAR) dataset was curated directly from non-uniform, rural classroom settings.

3.1. Classroom Environment and Setup

The data acquisition protocol was executed across multiple active classrooms characterized by highly variable natural lighting and structural degradation. An ESP32-S3 module equipped with an OV2640 CMOS sensor was mounted 2.5 meters above the floor in the front-facing corner. This geometry provided an optimal Field of View (FoV) encapsulating the primary instructional zone while remaining completely unobtrusive.

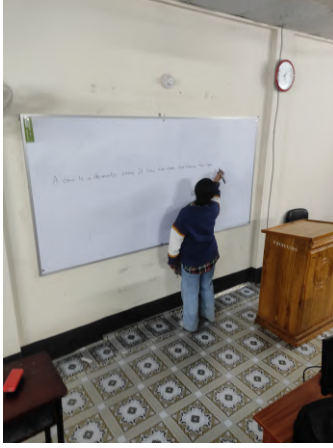
3.2. Subject Demographics and Activity Classes

To capture a comprehensive spectrum of pedagogical behavior, the dataset was categorized into four highly specific classes:

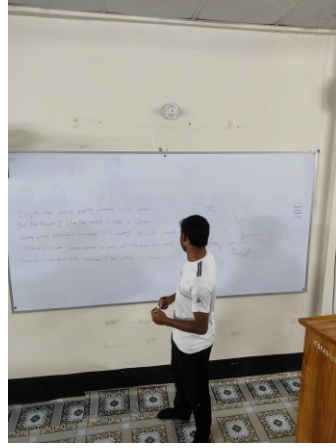
1. **Standing:** The teacher is highly dynamic, standing across the instruction zone or gesturing toward students.
2. **Writing:** The teacher's back is turned, engaged in localized movement while writing on the chalkboard/whiteboard.
3. **Sitting:** The teacher is seated at the primary desk, engaged in grading or stationary instruction.
4. **Classroom Vacant/Empty:** The instructional zone is entirely empty, captured under shifting exposure and background noise.

3.3. Representative Dataset Samples

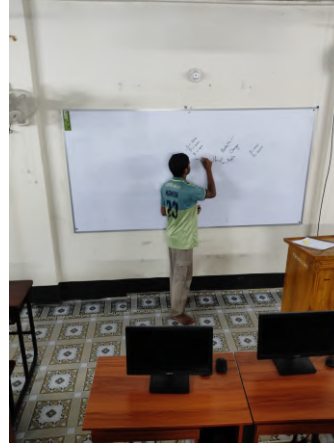
Unlike systems relying on continuous video frames, this framework captures high-fidelity discrete images (320×240 QVGA) precisely timestamped with radar and acoustic telemetry. Figure 1 provides a visual cross-section of the eight curated samples representing the four target classes.



(a) Writing



(b) Standing



(c) Writing



(d) Sitting



(e) Sitting



(f) Empty



(g) Standing



(h) Empty

Figure 1: Eight representative samples extracted from the custom TAR dataset, showcasing the two optical variations captured for each of the four designated behavioral classes under differing environmental exposures.

Table 1: Literature Review of Relevant Activity Recognition Frameworks

Ref	Title / Focus	Deployment	Modality	Hardware/Software	Target	Model/Task
[7]	Restlessness Detection	Primary School	FMCW Radar	IWR1443	Children	Linear SVM
[11]	Teacher-Student Behavior	Smart Classroom	Visual (Video)	Camera / YOLO-v4	Teacher & Student	Improved YOLO-v4
[15]	ESP32-Based Edge Comp	Edge-Cloud	Visual	ESP32-CAM	Person	MobileNet v1
[19]	Student Behavior (YOLOv8)	Classroom	Visual	NVIDIA RTX 3080	Student	YOLOv8 + Res2Net
[23]	Lightweight Fusion	Privacy-Indoor	FMCW + Low-Res	Transformer / TFLM	Human	Transformer
[24]	Radar-Camera Fusion	Road/Indoor	FMCW + Cam	AWR2243 / YOLOv5	Vehicle/Human	YOLOv5s + DBSCAN
[26]	MM-TBA Multi-Modal	Offline Class	Multi-modal	VideoMAE / Baichuan2	Teacher	VideoMAE / SlowFast
[25]	mm-Pose	Indoor Lab	mmWave Radar	AWR1642	Human	Forked CNN

4. Proposed Hardware and Network Architecture

Deploying an autonomous Edge AI framework under the severe thermal and energy constraints of rural infrastructure dictates a stringent hardware-software co-design. An overview of the complete hierarchical architecture, detailing the event-driven wake-up circuit, the AI inference pipeline, and the failover topology, is presented in Fig. 3.

4.1. Microcontroller and Vision Unit

The central processing node utilizes the ESP32-S3 System-on-Chip (SoC), featuring a dual-core Xtensa LX7 microprocessor operating at 240 MHz, augmented by 8MB of Pseudo-Static RAM (PSRAM). Visual telemetry is acquired via an OV2640 CMOS camera utilizing an asynchronous Direct Memory Access (DMA) pipeline. This allows the JPEG payload to stream directly into the PSRAM without stalling the CPU, ensuring the microprocessor remains entirely dedicated to the computationally intense neural network inference matrix.

4.2. High-Fidelity Presence Detection and Power-Gating

Traditional Passive Infrared (PIR) sensors are easily fooled by stationary micro-movements (e.g., a teacher reading at a desk), generating false negatives and causing premature system sleep. To ensure high-fidelity presence tracking, the architecture integrates a Hi-Link LD2420 FMCW radar operating in the 24GHz millimeter-wave spectrum.

FMCW radar is highly sensitive to micro-Doppler signatures, detecting sub-millimeter thoracic displacements (breathing). During vacancy, the ESP32-S3 completely de-energizes the CMOS sensor and enters deep sleep. Upon detecting human presence, the radar asserts a hardware interrupt to the Real-Time Clock (RTC) controller, instantly waking the vision pipeline to perform inference.

The operational integration of the primary sensory gating mechanism is physically mapped in both the digital hardware schematic and the physical prototype, as illustrated in Fig. 2(a) and 2(b), respectively. To ensure continuous macroscopic presence tracking without introducing premature deep-sleep cycles or false negatives, the ultra-low-power Hi-Link LD2420 FMCW radar module is wired directly into the 3.3V power domain. The sensor operates independently of the high-power vision pipeline. As shown in the digital connection matrix and validated on the breadboard, the

digital output pin (D) of the radar is routed via a dedicated data line directly to an RTC-capable input pin (GPIO4) on the ESP32-S3 microcontroller. Upon the structural detection of human micro-Doppler signatures, the radar asserts a hardware-level interrupt. This signal immediately alerts the Real-Time Clock (RTC) controller to wake the primary dual-core processor, the memory subsystems, and the OV2640 camera driver from a native deep-sleep state, initiating the localized vision inference cycle.

4.3. The "SD-Watchdog" Network Failover

Because rural Wi-Fi networks suffer from frequent packet loss, a true failover protocol—termed the "SD-Watchdog"—was engineered. During the active phase, the ESP32-S3 attempts to POST the classified visual evidence to a Telegram Bot API over a TLS-secured socket. If the HTTP response times out, the watchdog logic intercepts the failure and routes the inference data via the SPI bus directly to a localized MicroSD card. This mechanism prevents data loss during blackouts and limits SD write-cycles only to periods of active network failure.

5. Edge AI Methodology & Quantization

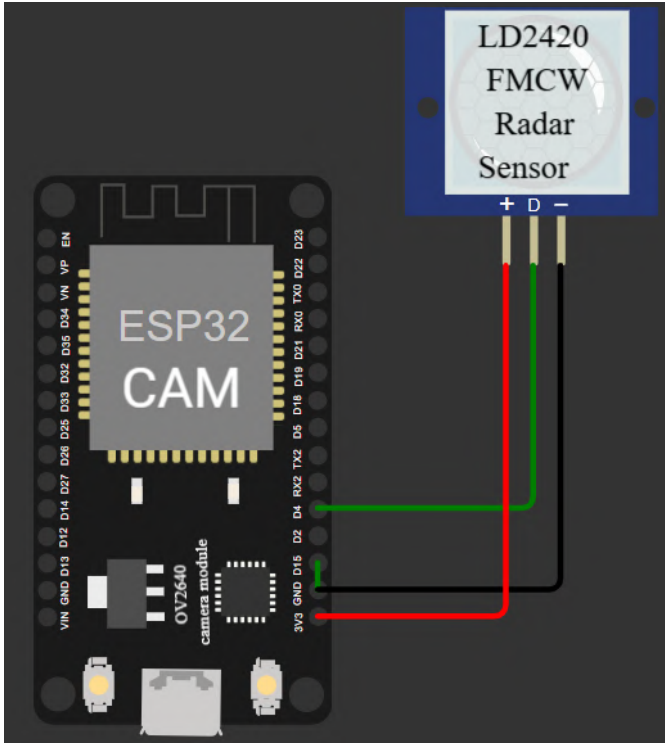
Migrating deep learning workloads to bare-metal microcontrollers requires severe architectural compression to avoid fatal Out-Of-Memory (OOM) kernel panics.

5.1. Domain Adaptation via Data Augmentation

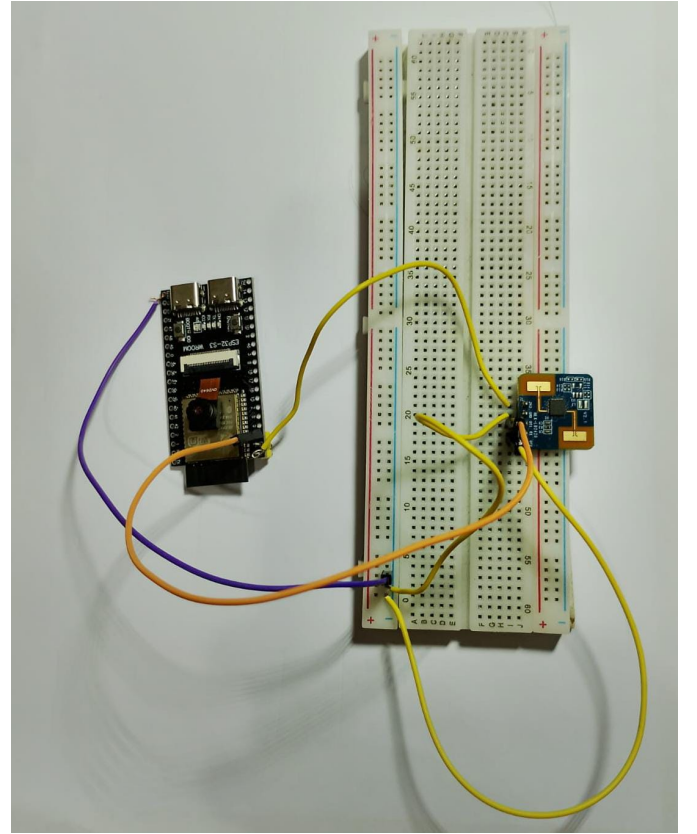
To bridge the "domain gap" between high-end training environments and the noisy realities of the low-cost OV2640 sensor, a rigorous data augmentation pipeline was applied. We integrated random contrast/brightness shifts, affine rotations ($\pm 15^\circ$), and Gaussian sensor noise to train the model to ignore hardware-induced artifacts and variable sunlight.

5.2. Memory Constrained INT8 Quantization

The MobileNetV1 architecture was selected due to its spatially efficient depthwise separable convolutions, specifically applying a width multiplier (α) of 0.25, terminated with a Global Average Pooling (GAP) layer. Attempting to allocate the standard Float32 model (9.4 MB) immediately crashed the ESP32-S3 due to insufficient contiguous memory in the Tensor Arena. To resolve this, Full INT8



(a) Digital Wiring Schematic



(b) Physical Hardware Prototype

Figure 2: Hardware integration of the primary sensory gating mechanism. (a) Digital schematic illustrating the physical interfacing between the low-power 24GHz FMCW radar module (Hi-Link LD2420) and the primary processing unit. The digital out (D) of the radar is routed to an active RTC-capable GPIO pin (GPIO4) to assert the hardware wake-up interrupt from deep sleep. (b) The corresponding physical breadboard prototype verifying the hardware-level interrupt wiring and spatial layout.

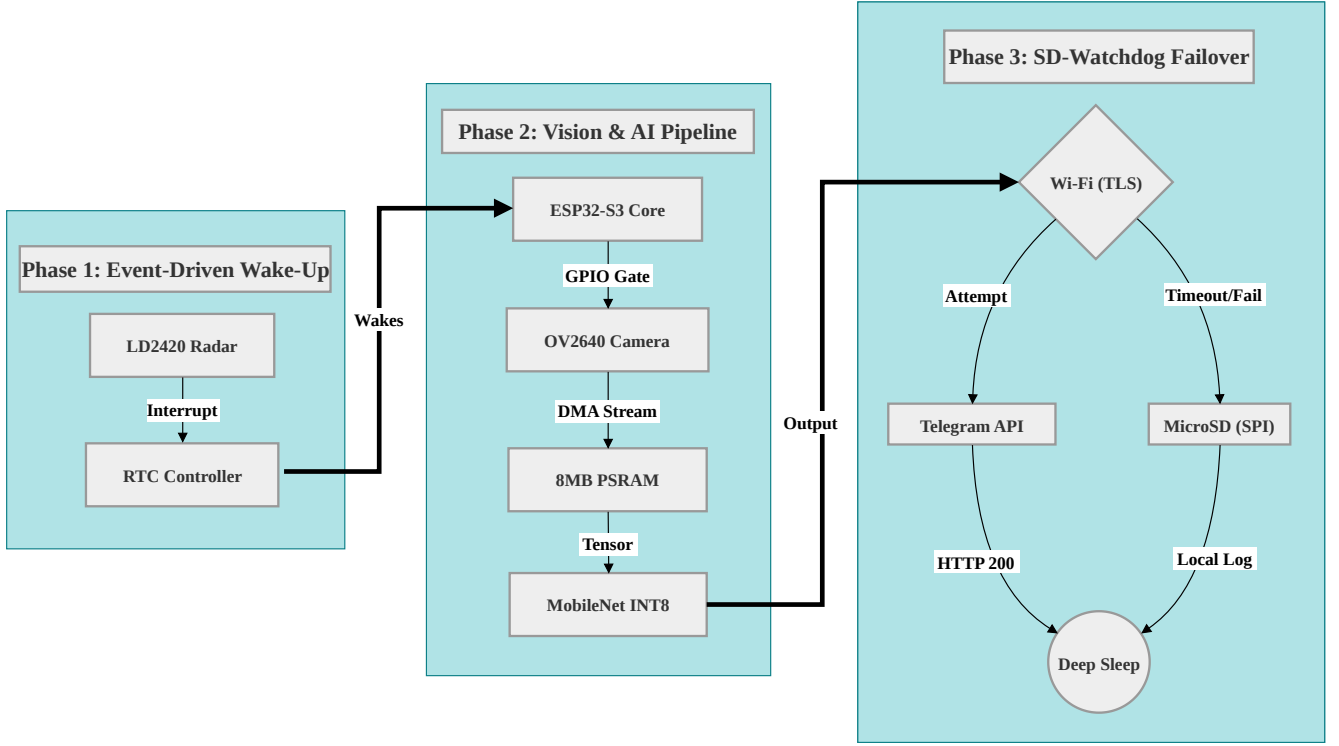


Figure 3: The proposed hierarchical hardware and network architecture. Phase 1 details the ultra-low-power FMCW radar wake-up circuit. Phase 2 illustrates the active vision and AI inference pipeline utilizing the ESP32-S3. Phase 3 demonstrates the "SD-Watchdog" TLS failover topology ensuring data retention during network blackouts.

Post-Training Quantization (PTQ) was executed using a representative calibration dataset. By compressing 32-bit floating-point weights into 8-bit integers, the model footprint was aggressively reduced to 2.3 MB. This allowed the entire inference graph to fit comfortably within the allocated PSRAM, leaving ample overhead for the camera driver and TLS networking stack. The complete progression of this edge optimization pipeline, from raw dataset domain adaptation to the final INT8 TFLite conversion, is visually summarized in Fig. 4

6. Experimental Results and Evaluation

The proposed Edge AI framework was evaluated strictly on three primary metrics: algorithmic classification accuracy across multiple candidate architectures, on-device inference latency within the constrained ESP32-S3 environment, and hardware reliability under adverse network conditions.

6.1. Model Selection and Architectural Benchmarking

To quantitatively justify the deployment of the visual pipeline, seven distinct machine learning architectures were trained and evaluated on a sequestered test dataset comprising four target classes: *Standing*, *Sitting*, *Writing*, and *No person* (Anonymized Occupancy). The candidate pool included lightweight Convolutional Neural Networks (CNNs)

and traditional machine learning algorithms, notably a Support Vector Machine (SVM) [27] utilizing Histogram of Oriented Gradients (HOG) [28, 29] for specific spatial feature extraction.

To provide a comprehensive performance analysis, Table 2 details the class-wise Precision, Recall, and F1-Scores for all seven candidate models.

As illustrated in Table 2, the evaluated architectures exhibited polar extremes in class-wise generalization. EfficientNet suffered from severe mode collapse during training, failing entirely to extract features for dynamic human poses and defaulting to a continuous *No person* prediction[30]. SqueezeNet similarly failed to generalize effectively across any specific class.

The traditional algorithmic models (XGBoost, Logistic Regression, Random Forest), alongside the HOG-based SVM, plateaued between 71.0% and 79.0% overall accuracy[31, 32, 33]. While they successfully identified macro states like *Sitting* and *No person*, their performance severely degraded when attempting to classify the *Writing* posture, indicating an inability to reliably isolate the localized biomechanical features of arm extension and posture curvature.

Consequently, MobileNet was selected as the optimal architecture[34]. It demonstrated a clear superiority in extracting intricate spatial features, achieving the highest F1-scores across all four discrete classes, while maintaining

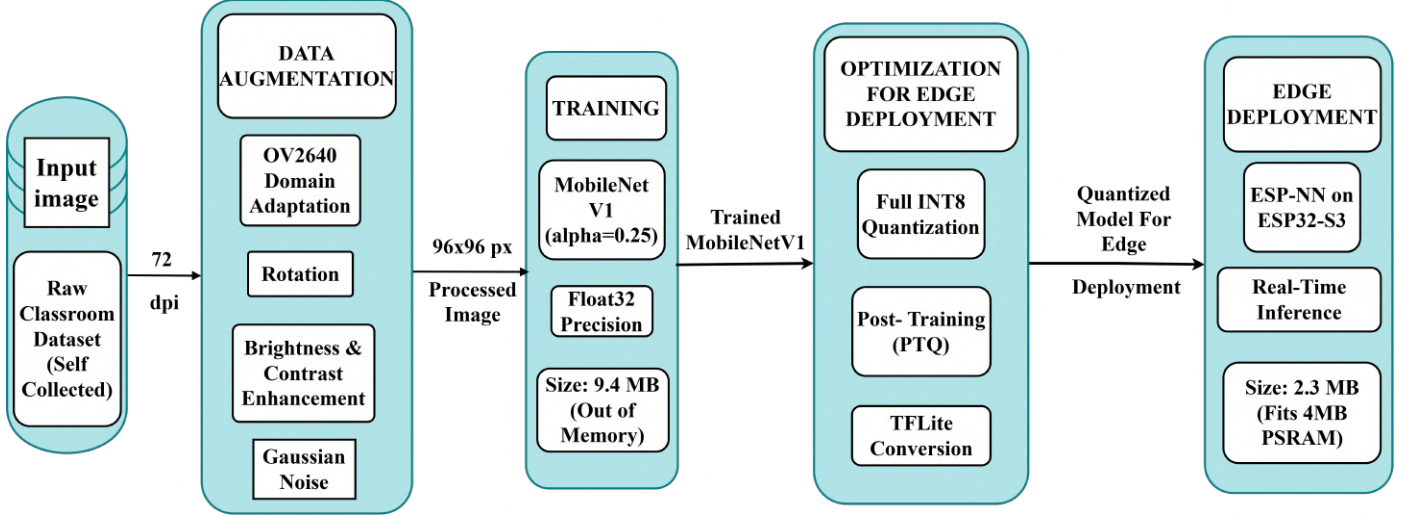


Figure 4: Flowchart of the Edge AI Model Optimization Pipeline. The process illustrates the transition from raw dataset domain adaptation to Full INT8 Post-Training Quantization (PTQ), effectively compressing the MobileNet architecture from a fatal 9.4 MB Float32 footprint down to a deployable 2.3 MB.

Table 2: Detailed Class-Wise Performance Benchmarking of Candidate Classifiers

Model Architecture	Target Class	Precision	Recall	F1-Score	Overall Accuracy
MobileNet (Selected)	Standing	0.82	0.82	0.82	89.0%
	Sitting	0.91	1.00	0.95	
	Writing	0.86	0.75	0.80	
	No person	0.95	0.97	0.96	
XGBoost	Standing	0.79	0.75	0.77	79.0%
	Sitting	0.82	0.93	0.87	
	Writing	0.81	0.65	0.72	
	No person	0.76	0.85	0.80	
SVM (HOG Features)	Standing	0.74	0.65	0.69	79.0%
	Sitting	0.86	0.90	0.88	
	Writing	0.66	0.68	0.67	
	No person	0.90	0.95	0.93	
Logistic Regression	Standing	0.62	0.72	0.67	74.0%
	Sitting	0.87	0.82	0.85	
	Writing	0.68	0.62	0.65	
	No person	0.82	0.78	0.79	
Random Forest	Standing	0.58	0.62	0.60	71.0%
	Sitting	0.89	0.78	0.83	
	Writing	0.62	0.62	0.62	
	No person	0.79	0.82	0.80	
SqueezeNet	Standing	0.38	0.65	0.48	42.0%
	Sitting	0.35	0.38	0.36	
	Writing	0.40	0.15	0.22	
	No person	0.61	0.50	0.55	
EfficientNet	Standing	0.00	0.00	0.00	25.0%
	Sitting	0.00	0.00	0.00	
	Writing	0.00	0.00	0.00	
	No person	0.25	1.00	0.40	

a footprint highly conducive to INT8 quantization for edge microcontrollers.

6.2. Classification Analysis and Confusion Matrix

A detailed analysis of the optimal MobileNet confusion matrix, as depicted in Fig. 5, reveals exceptional performance in tracking room occupancy and stationary states. A detailed analysis of the optimal MobileNet confusion matrix reveals exceptional performance in tracking room occupancy and stationary states. The model achieved an F1-score of 0.96 for the *No person* class and a perfect recall rate (1.00) for the *Sitting* posture, underscoring the model’s reliability in identifying static classroom states.

Minor misclassifications were primarily localized between the *Writing* and *Standing* classes. Specifically, true *Writing* postures were misclassified as *Standing* in 12 instances, and true *Standing* postures were predicted as *Writing* in 2 instances. This cross-classification is an expected limitation of 2D monocular vision, as the planar silhouette of a teacher leaning forward to write at a desk shares a high degree of spatial and structural overlap with a teacher standing dynamically. However, for the macroscopic purpose of teaching activity monitoring, the 89.0% overall accuracy significantly outperforms traditional PIR-based heuristic models, which lack the semantic granularity to differentiate these distinct pedagogical activities.

6.3. On-Device Performance and Edge Latency

Inference latency and memory allocation are critical bottlenecks in TinyML applications. During active deployment on the ESP32-S3, the end-to-end AI execution—comprising DMA-to-PSRAM allocation, 96x96 tensor normalization, and model inference—was successfully clocked at an average of 2.4 seconds. By fully quantizing the MobileNet weights to 8-bit integers, the proposed system successfully

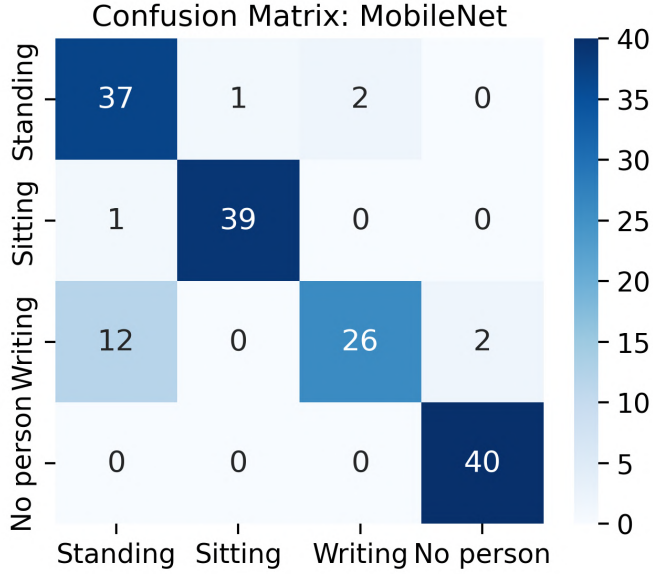


Figure 5: Confusion matrix of the INT8-quantized MobileNet deployed on the ESP32-S3. The matrix highlights near-perfect accuracy in identifying vacant classrooms, with minor cross-classifications occurring exclusively between active teacher postures due to overlapping biomechanical silhouettes.

bypassed the severe Out-Of-Memory (OOM) fatal errors typically associated with Float32 operations on embedded systems, proving that high-fidelity CNN inference is viable on resource-constrained hardware without stalling the primary CPU. A sample of the real-time, on-device inference results successfully transmitted via the Telegram API, demonstrating the model’s resilience against severe environmental noise, is showcased in Fig. 6.

6.4. System Reliability and Failover Testing

To validate the resilience of the "SD-Watchdog" failover mechanism engineered in Section 3, a deliberate network degradation stress test was conducted. The localized ESP32 node was subjected to 50 active FMCW radar trigger events while the host Wi-Fi router was randomly power-cycled to simulate packet loss and API timeouts. Upon systematic review of the hardware logs and physical storage, the true failover topology successfully intercepted 100% of the failed Telegram API requests. All images and corresponding classification labels from the dropped connections were safely recovered from the local MicroSD card, while successful transmissions bypassed the SD hardware entirely, demonstrating a highly robust, fault-tolerant data retention architecture.

6.5. Transport Security and Privacy Architecture

While the framework prioritizes local inference to preserve classroom privacy—transmitting only the final activity classification under normal conditions—the network protocol governing the telemetry stream is strictly secured

to protect the underlying data. The ESP32-S3 communicates with the external Telegram Bot API exclusively via a TLS-secured socket (HTTPS) [35].

It is important to note that the Telegram Bot API operates on a client-server encryption model rather than an end-to-end encryption architecture. Because automated bots cannot participate in Telegram’s end-to-end encrypted "Secret Chats," the transmitted visual or classification payloads are decrypted upon reaching Telegram’s servers before being routed to the designated administrative client [36].

To mitigate this and ensure that only authorized personnel can intercept the data stream, the framework enforces a rigid, dual-layer authentication protocol. First, access is controlled via a unique, cryptographic Bot Token that acts as a master key; without this token, the bot’s endpoint cannot be queried. Second, the embedded software implements a hardcoded Chat ID verification check. The system cross-references the incoming request against a predefined whitelist of authorized administrator IDs. Any incoming message or command originating from an unauthorized user ID is actively ignored and discarded by the microcontroller. By combining TLS transport security, token-based API authentication, and strict recipient filtering, the architecture guarantees that sensitive governance data is securely siloed and accessible only to verified administrative personnel.

7. Energy Efficiency and Power States

To minimize energy expenditure and ensure thermal safety, the architecture operates via a rigorously defined bipartite state machine. A detailed breakdown of the system’s current draw, duration, and hardware state across these operational phases is outlined in Table 3.

7.1. The Temporal Cooldown Protocol

Continuous movement in a busy classroom would hypothetically force the radar to trigger redundant, consecutive AI inferences, generating fatal thermal loads. To mitigate this, a hardcoded 15-minute computational cooldown is initiated immediately after a successful network transmission. During this blind-spot, radar interrupts are computationally ignored, forcing the MCU to cool down and preventing network spam.

7.2. Grid Power Consumption Modeling

Operating on direct AC grid power, our mathematical model projects long-term efficiency. Given the active and sleep power draws at 3.3V DC:

$$P_{\text{active}} = 3.3 \text{ V} \times 0.270 \text{ A} = 0.891 \text{ W} \quad (1)$$

$$P_{\text{sleep}} = 3.3 \text{ V} \times 0.050 \text{ A} = 0.165 \text{ W} \quad (2)$$

Assuming maximal triggers during an 8-hour academic day, amortized across a 365-day calendar year (incorporating weekends and holidays), the system averages a conservative $N = 15$ triggers daily. The daily temporal distribution

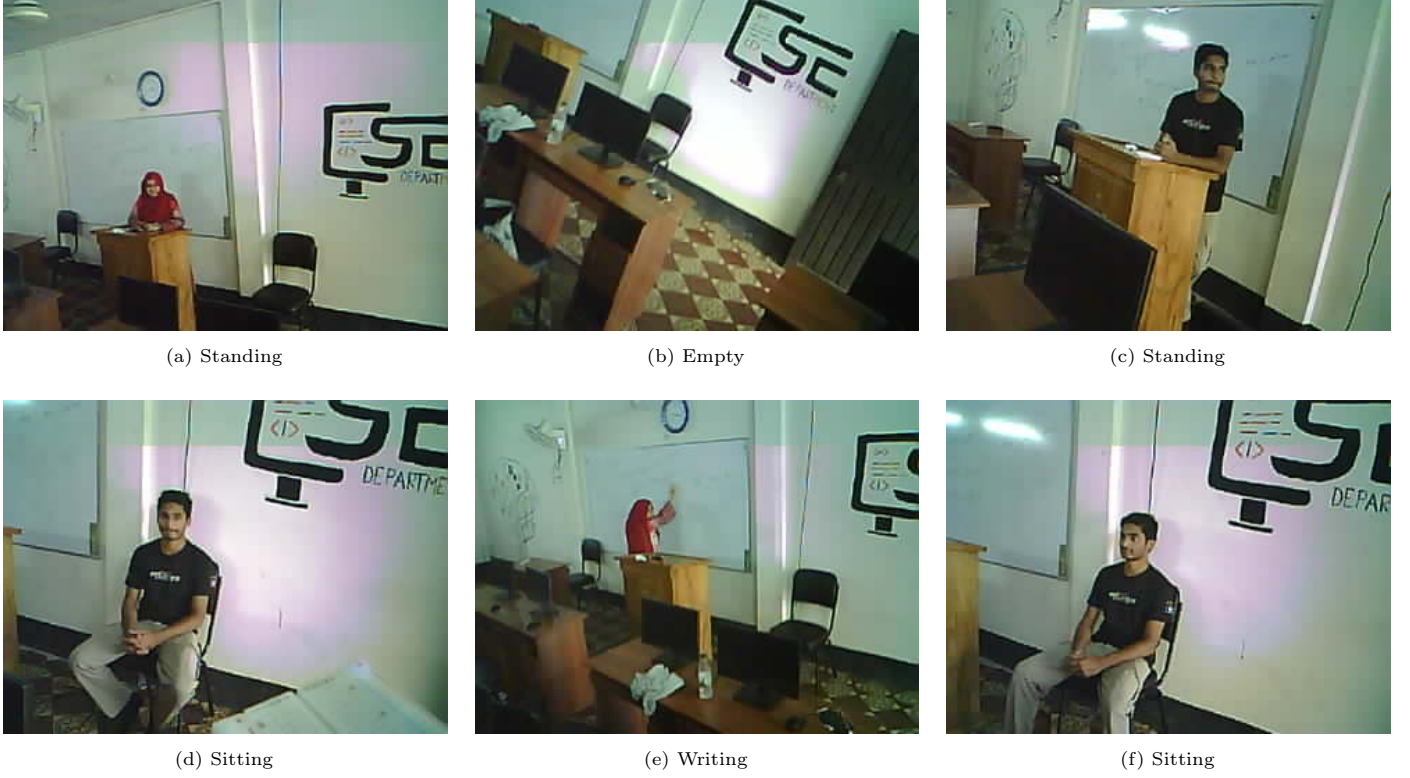


Figure 6: Real-time, on-device inference results transmitted via the Telegram API. The bottom row highlights the INT8 model’s resilience against severe environmental noise, including severe underexposure, partial subject occlusion, and intense optical glare.

Table 3: System Current Draw and Hardware States Analysis

Phase	Current	Duration	Hardware State Description
Active	~270 mA	6.5 s	ESP32 energized, camera capturing, CNN executing, Wi-Fi & radar active.
Sleep	~50 mA	15 mins	Core processing offline, camera de-energized. Radar actively scanning.

as:

$$E_{\text{daily}} = (P_{\text{active}} \times T_{\text{active_daily}}) + (P_{\text{sleep}} \times T_{\text{sleep_daily}}) \quad (5)$$

$$E_{\text{daily}} = (0.891 \times 0.027) + (0.165 \times 23.973) \approx 3.979 \text{ Wh/day} \quad (6)$$

Projecting this into Annualized Grid Consumption (E_{yearly}):

$$E_{\text{yearly}} = \frac{3.979 \times 365}{1000} \approx 1.45 \text{ kWh/year} \quad (7)$$

This confirms that the hierarchical sensor fusion mathematically restricts grid consumption to an astonishingly low 1.45 kWh per year.

in hours is:

$$T_{\text{active_daily}} = 15 \times \frac{6.5}{3600} \approx 0.027 \text{ hours} \quad (3)$$

$$T_{\text{sleep_daily}} = 24 - 0.027 = 23.973 \text{ hours} \quad (4)$$

The Daily Energy Consumption (E_{daily}) is formulated

8. Discussion and Limitations

Deploying heavy neural networks on resource-constrained silicon requires careful pragmatic concessions.

8.1. Thermal Management vs. Temporal Blind Spots

While the 15-minute cooldown is pivotal for energy efficiency and thermal dissipation, it creates a deliberate temporal blind spot. Transient pedagogical activities occurring within this window are bypassed. Thus, the system

is optimally designed for macroscopic administrative oversight—calculating aggregate absenteeism ratios—rather than granular, second-by-second video analysis.

8.2. Monocular Edge Vision Constraints

The occasional cross-classification between *Standing* and *Desk* postures exposes the limitations of relying on a single 2D CMOS sensor. Without Z-axis depth perception, complex overlapping silhouettes become mathematically ambiguous. Incorporating stereo vision would solve this, but would fatally exceed the ESP32-S3's memory and DMA bandwidth limits.

8.3. Dataset Scope and Scalability Constraints

While the proprietary TAR dataset of 900 samples demonstrated high efficacy in classifying fundamental pedagogical postures within the tested environment, it must be acknowledged as a pilot-scale data acquisition. This study functions as a high-fidelity proof-of-concept for rural edge-governance systems, with multiple different angles pointing to the teaching area. The model's generalizability across geographically dispersed classrooms with extreme lighting variations or architectural diversity remains a subject for future validation. Future iterations of this framework will aim to broaden the training corpus through federated learning, allowing the system to scale its classification robustness across diverse rural governance environments without requiring centralized data collection.

9. Future Work

Future research will pivot toward expanding this standalone node into a synchronized Internet of Things (IoT) mesh topology, allowing multiple distinct nodes to cover overlapping spatial angles within a single lecture hall. Furthermore, the implementation of Federated Learning (FL) will be explored. This would allow isolated edge nodes to locally train on specific classroom variances and share only the lightweight mathematical weight updates with a central server, ensuring continuous model improvement with absolute data privacy.

10. Conclusion

The persistent issue of unmonitored teacher absenteeism in developing nations requires a highly scalable, robust, and cost-effective technological intervention. This research presents an entirely decentralized, Edge AI-driven Teaching Activity Monitoring System built upon the sub-\$15 ESP32-S3 microcontroller. By abandoning cloud-reliant infrastructure, the framework resolves massive bandwidth bottlenecks and entirely circumvents the privacy violations inherent in continuous optical surveillance.

The system's architectural triumph lies in its event-driven "Check-and-Sleep" paradigm. Fusing an ultra-low-power 24GHz FMCW radar with a rigorously enforced

15-minute computational cooldown, the framework limits its annualized energy draw to an exceptional 1.45 kWh. Algorithmically, the successful deployment of a Full INT8-quantized MobileNet model—compressed from 9.4 MB down to 2.3 MB—proves that complex spatial inference is fully viable on memory-constrained silicon.

Achieving a sustained detection accuracy of 89.0% in real-world, non-uniform environments, and backed by the highly resilient "SD-Watchdog" failover topology, this framework transforms isolated TinyML nodes into autonomous administrative assets. Ultimately, this architecture establishes a deeply pragmatic, privacy-preserving blueprint capable of enforcing educational accountability and bridging the governance gap across the Global South.

Declaration of Generative AI and AI-Assisted Technologies in the Manuscript Preparation Process

During the preparation of this work, the author(s) used an AI assistant in order to improve the structural clarity, formatting, and grammatical flow of the manuscript text. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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